UTM UNIVERSITI TEKNOLOGI MALAYSIA | Faculty of Computing

SECR3443 - INTRODUCTION TO CRYPTOGRAPHY

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Project Documentation : CryptoDetective

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Video and Code File Link: BADZ_CryptoDetective

Prototype Link: https://dnialhkm.github.io/CryptoProject/

TABLE OF CONTENT

1.0 Problem Statement	2
2.0 Target Users	2
3.0 Learning Outcomes	3
4.0 App/game Concept	4
5.0 AI Integration	4
6.0 Tools Used	4
7.0 Game/app Flow	5
8.0 Cryptography Explanations	6
9.0 AI Feature Designs	7
10.0 Screenshots or Mock-ups	8
11.0 User Testing	12
12.0 Individual Reflections	14

1.0 Problem Statement

Cryptography plays a crucial role in modern digital communication, cybersecurity and information integrity. However, most secondary school students (ages 13–17) have limited exposure to cryptography concepts, and existing learning materials tend to be too technical, abstract, or math-heavy. As a result, students struggle to understand:

- What encryption and decryption mean,
- How hashing ensures data integrity,
- The purpose of digital signatures, and
- Why message authenticity matters in real-world scenarios.

There is a need for an educational tool that introduces cryptography concepts in a simple, engaging, and interactive way without requiring advanced mathematical background.

2.0 Target Users

The primary target users are:

- Secondary school students aged 13–17
- With minimal or no prior cryptography knowledge
- Who are familiar with basic computer use and digital learning environments.

User characteristics include:

- Curious and problem-solving oriented
- Prefer engaging, story-based learning rather than theoretical explanations
- Benefit from visual cues and guided assistance

The application assumes no background in cybersecurity, networking, or programming, ensuring the content remains accessible and age-appropriate.

3.0 Learning Outcomes

By interacting with our proposed educational application, students will be able to:

Knowledge & Understanding

1. Describe the purpose of cryptography in digital communication.
2. Differentiate between encryption, hashing, and digital signatures at a conceptual level.
3. Understand the idea of message authenticity and tampering.

Practical Cognitive Skills

1. Analyze simple cryptographic evidence (hash values, signatures, encrypted messages).
2. Identify whether a message is authentic or tampered based on provided clues.

Digital & Problem-Solving Skills

1. Use guided hints to improve decision-making.
2. Strengthen reasoning by interpreting cryptographic clues rather than guessing.
3. Build awareness of real-world cybersecurity problems (e.g., fake messages, message interception).

Ethical Awareness

1. Recognize how cryptography protects privacy, integrity, and digital trust.
2. Appreciate the importance of ethical use of cryptographic technologies.

4.0 App/game Concept

CryptoDetective is a puzzle-based educational game designed to teach secondary school students the basic concepts of cryptography in a fun and interactive way. Players take on the role of a digital detective, investigating suspicious messages to determine whether they are authentic or tampered.

5.0 AI Integration

The AI integration that will be used in the system is an adaptive hint based on student mistakes. This feature is used when a student selects an incorrect option, the AI provides a hint that explains the concept clearly, without giving the answer directly. The AI integration allows students to quickly learn why an answer is incorrect and can adjust their reasoning. It also provides guidance to analyze the evidence and make informed decisions rather than guessing. Ultimately, the hints can help the students overcome challenges without revealing answers, while keeping them motivated and focused.

Some of the limitations of using AI for this system are that the AI only provides predefined hints and cannot perform real cryptographic calculations. This means it cannot verify messages or generate cryptographic results in real time. Next, the feedback given by the AI is conceptual and only focusing on explaining the reasoning behind each answer. This is to enable the students to still apply their own critical thinking to interpret clues and make decisions, rather than relying solely on AI guidance. Lastly, the risk in using AI is clearly that students may become overly dependent on the hints, which could reduce independent problem-solving skills if they do not actively analyze the evidence themselves.

6.0 Tools Used

The tool that we used for CryptoDetective is a coding-based prototype which is based on HTML, JavaScript and CSS. The prototype will be developed as a web application.

7.0 Game/app Flow

The game consists of three unique cases to be investigated. In each case students need to solve the mystery based on their cryptography knowledge. While solving the mysteries, AI will act as a detective assistant which will guide the student in simple terms when making mistakes. The flow of the game is as below.

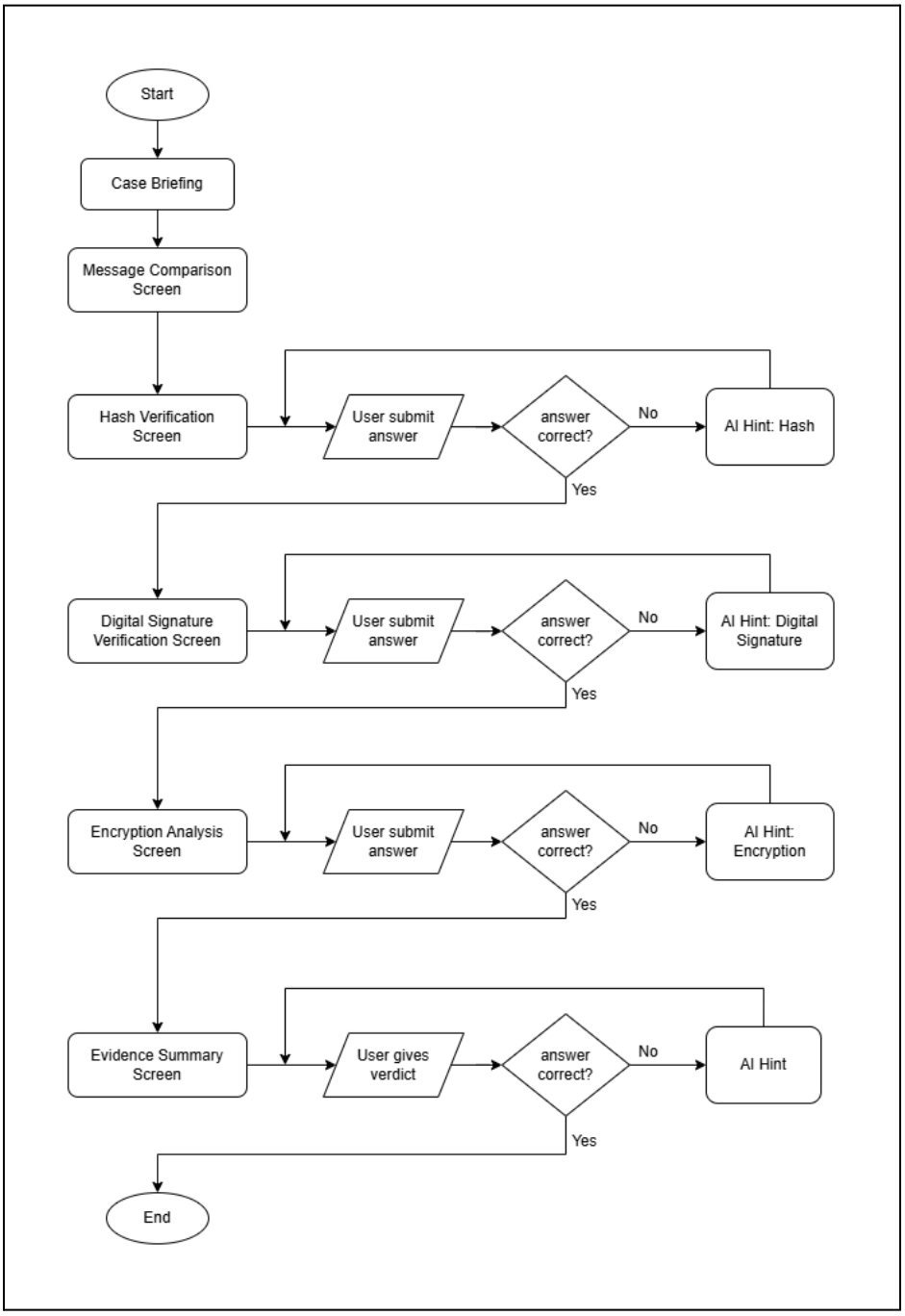

Figure 1: CryptoDetective Game Flow

8.0 Cryptography Explanations

CryptoDetective is a game to test students' cryptographic understanding of hashing, digital signatures, and encryption and decryption. Students must analyze case-based mysteries and solve them using their cryptographic knowledge.

Hashing is used to check whether a message has been tampered. A hash means a short digital fingerprint created from a message. That means the same message always produces the same hash and this process is a one-way operation. Even if one letter changes, the hash will become completely different. Students need to compare the original hash with the received hash to detect if a suspect has altered the message.

Next, a digital signature is used to prove a person's identity where they authenticate the sender of the message and determine the integrity of the message. In CryptoDetective, the game simply explains the idea of digital signatures where students must verify the digital signature to see if the message truly came from a trusted or fake sender.

Lastly, the concept of encryption is to change a readable message called plaintext into a secret form also called ciphertext. This is to ensure that even if someone steals the message, they will not be able to understand it. Whereas decryption simply means the opposite process where it returns the ciphertext into its original and readable form but only using the correct key. CryptoDetective allows students to analyze the encryption and decryption processes and also increase their awareness of encryption to help them understand message protection.

All in all, CryptoDetective helps the students to learn how modern systems protect digital messages. They will analyze the crime evidence and decide whether the message is authentic, the sender is trustworthy and whether the message has been intercepted or tampered with.

9.0 AI Feature Designs

The AI Detective Assistant in CryptoDetective is a smart helper designed to guide students like a tutor. Unlike complex AI tools that write new text every time, this system follows a strict set of rules. It uses a Condition-Action system, which means it has a list of pre-written hints stored in the game's code. The AI watches the player's choices. If the player clicks the wrong button, for example, saying two different hashes are matches, the AI instantly recognizes the mistake and picks the specific hint meant for that error. This ensures the advice is always 100% accurate and relevant to what the student is doing.

The primary goal of AI is to assist students in learning by thinking, rather than simply providing answers. This is called guided support. For instance, if a student struggles with a hashing puzzle, the AI won't just say "The answer is No." Instead, it will give a clue like, "Look closely at every character." This forces the student to look at the evidence again and figure out the solution themselves. By doing this, the AI helps fix misunderstandings about concepts like encryption and digital signatures without doing the work for the student.

In terms of user experience, the AI follows a Hidden until Needed rule design pattern to maintain a clean and focused interface. Most of the time, the AI assistant is invisible, so the student can solely focus on the game. It only pops up on the screen when the student makes a mistake. As soon as the student fixes their error and chooses the right answer, the AI box disappears. This gives the student immediate feedback, which is seeing the AI leave means they have successfully understood the concept and solved that part of the case.

10.0 Screenshots or Mock-ups

The figures below are the screenshots of the prototype.

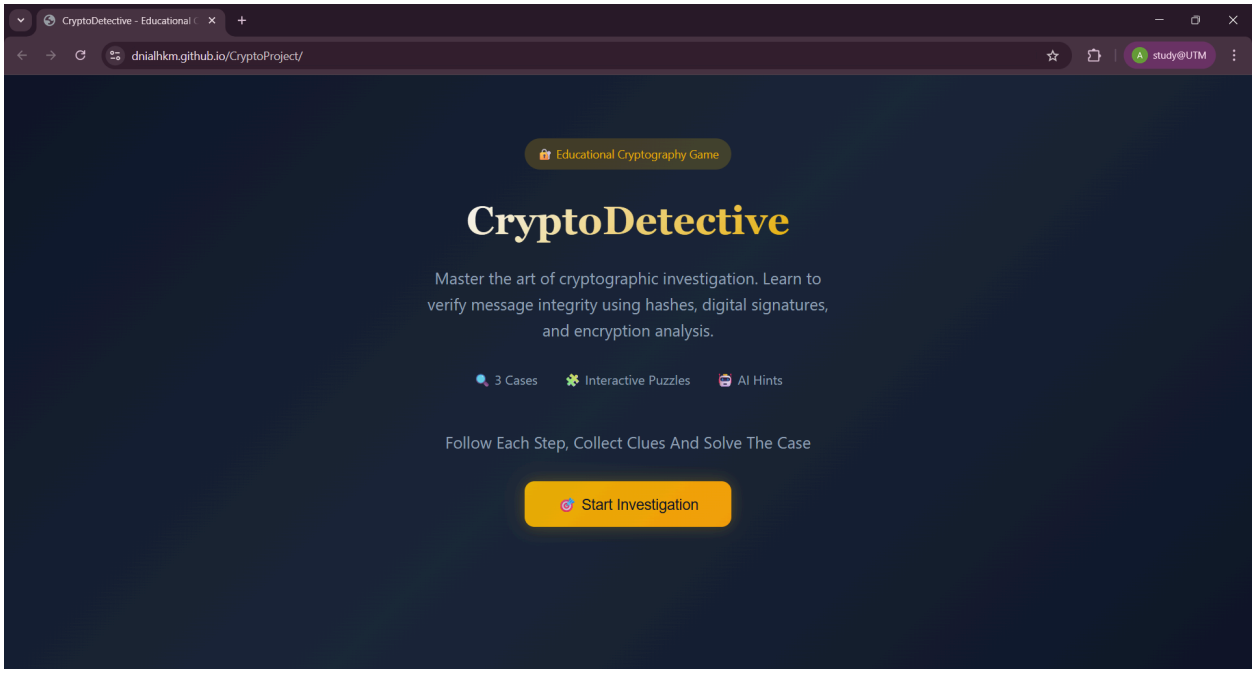

Figure 2: Main Page of CryptoDetective

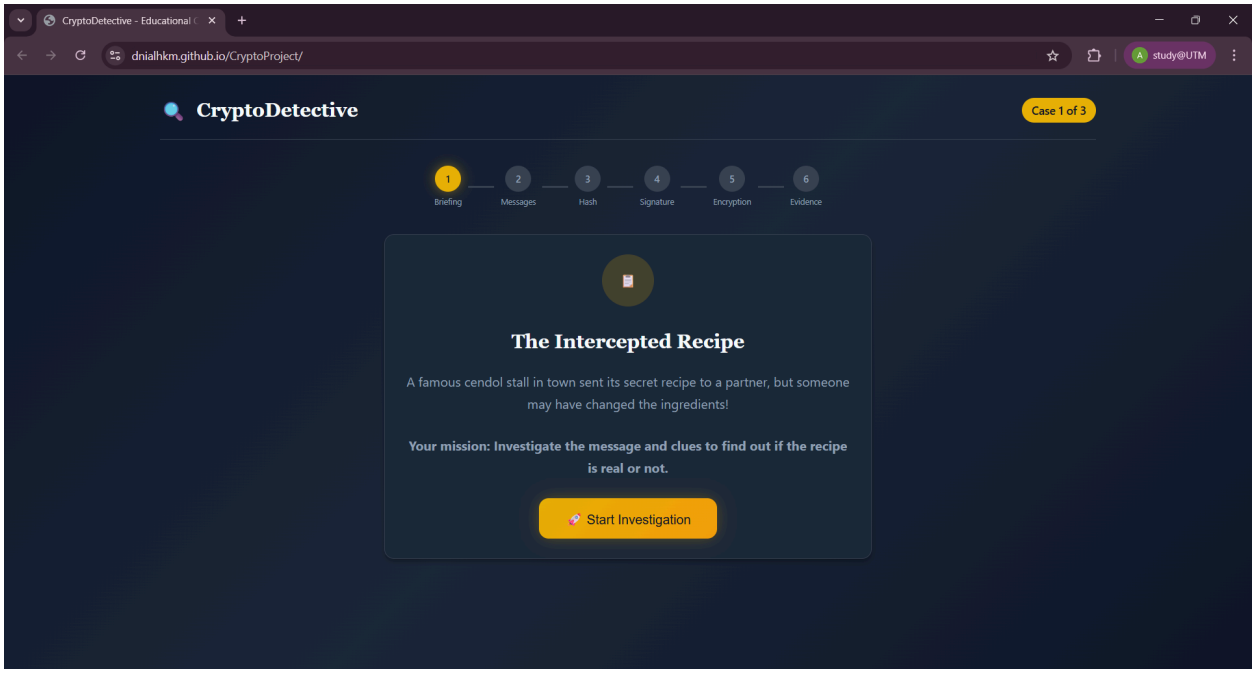

Figure 3: Mission briefing interface

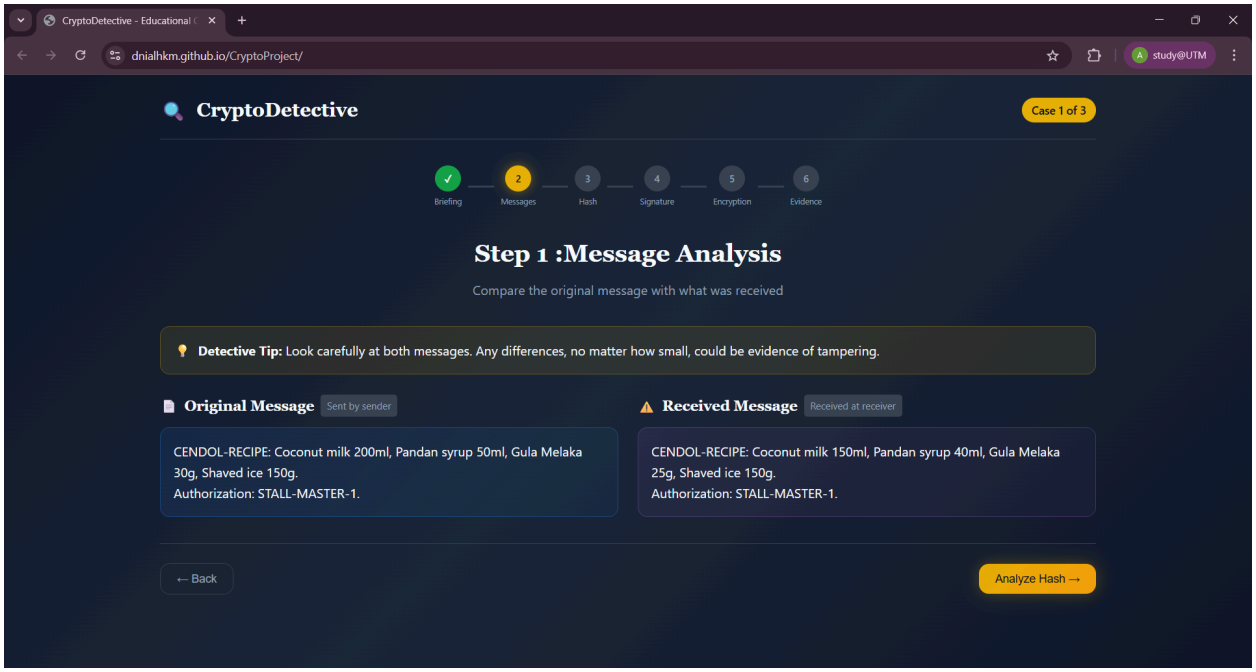

Figure 4: Message display and analysis interface

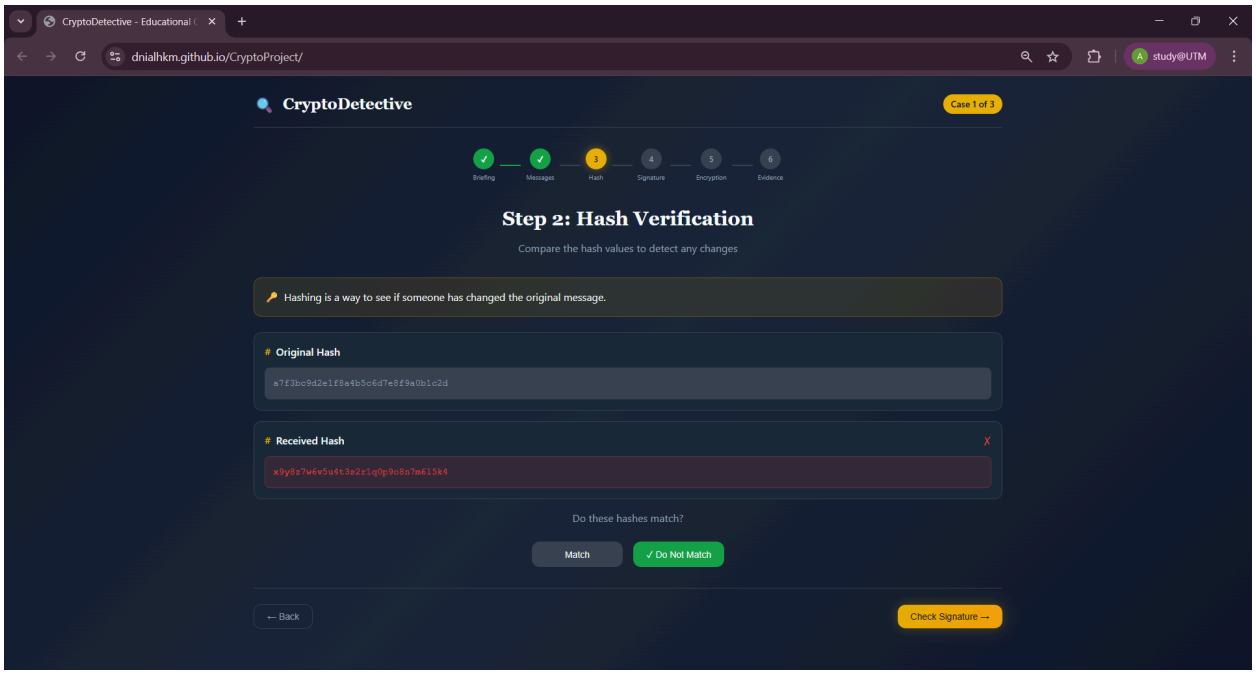

Figure 5: Hash verification interface

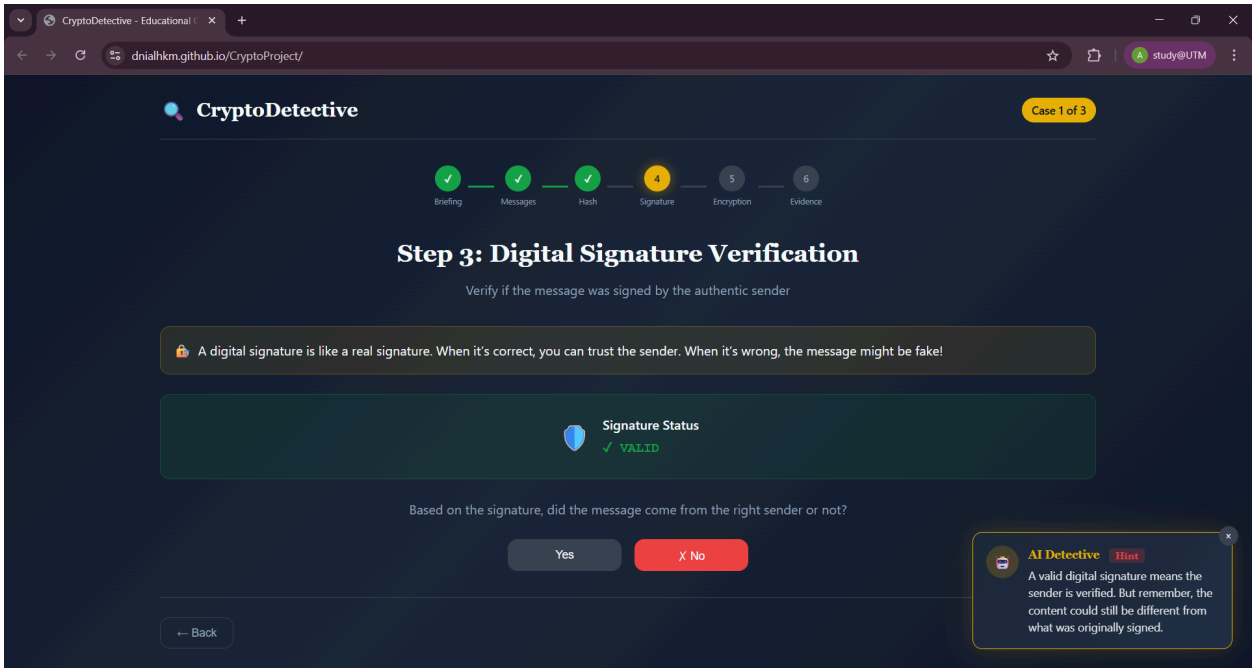

Figure 6: Digital Signature Verification interface with AI hint

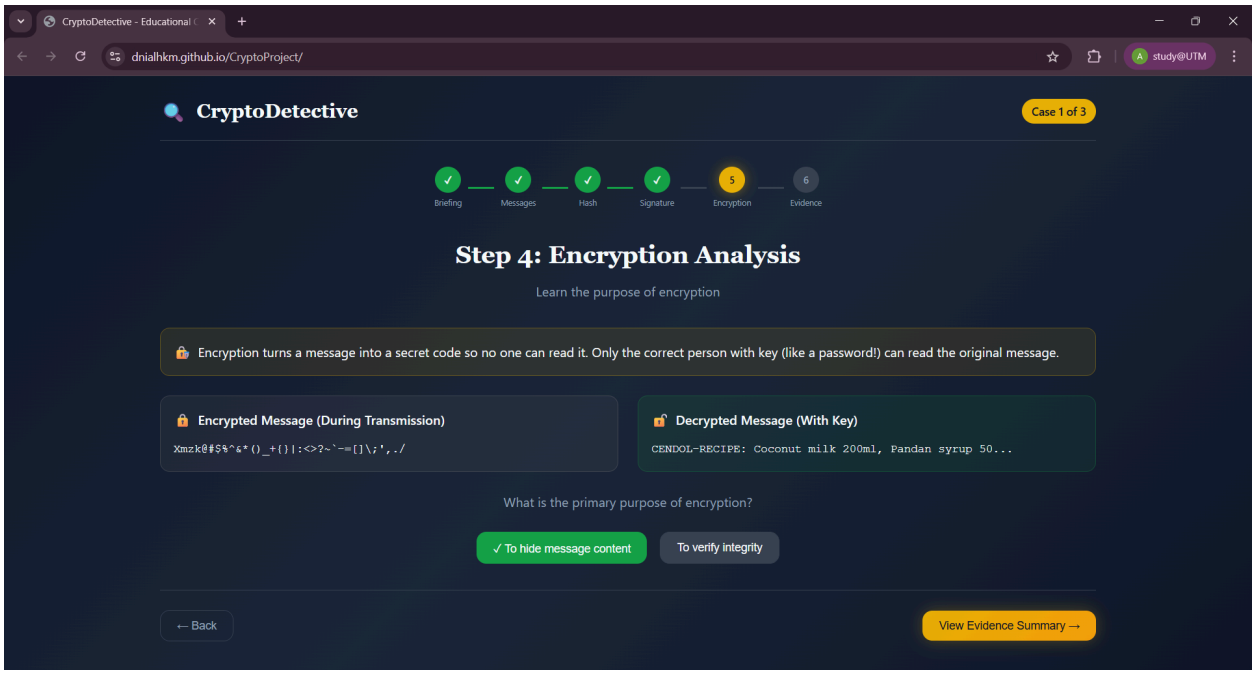

Figure 7: Encryption Analysis interface

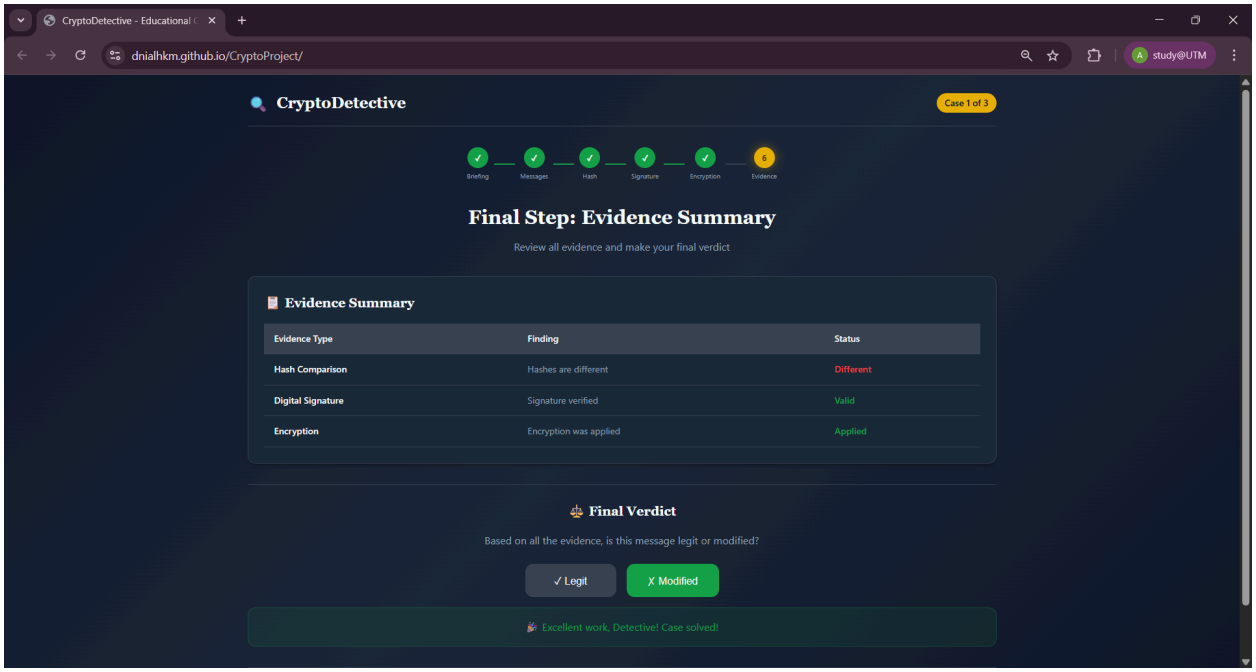

Figure 8: Evidence summary interface

11.0 User Testing

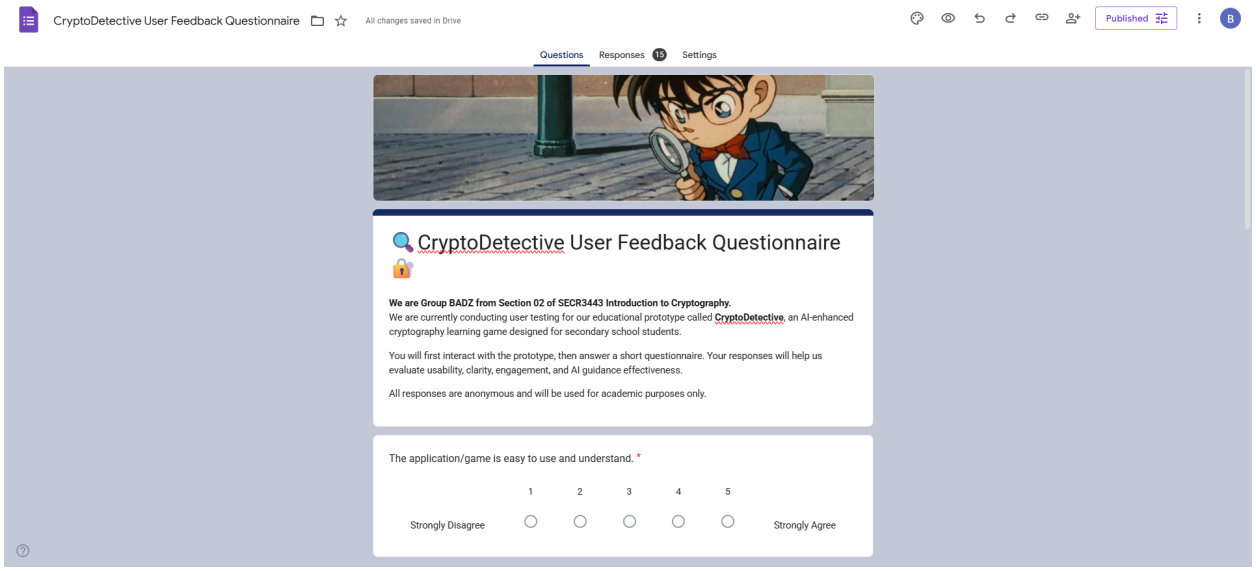

Figure 9: Cryptodetective Feedback Survey

User testing was conducted with fifteen (15) secondary school students to evaluate the usability, clarity, engagement level, and educational effectiveness of the CryptoDetective prototype. Participants interacted with the application and subsequently completed a structured questionnaire consisting of six likert scale items and two open-ended questions.

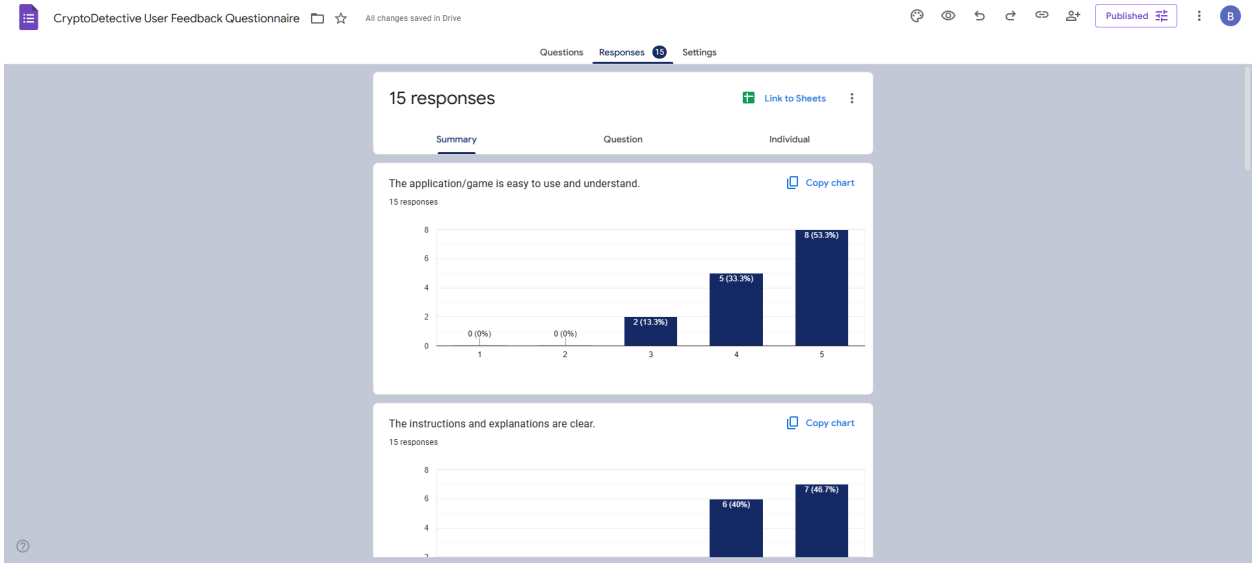

Figure 10: Quantitative Analysis of CryptoDetective

Based on the likert-scale responses, participants generally found the game easy to use and navigate, with clear instructions and understandable explanations. On average, the application scored 4.5 out of 5 for ease of use and 4.3 out of 5 for clarity of instructions. Students reported that the cryptography concepts were easier to understand after using the game, with an average score of 4.4, and they found the activities and gameplay engaging, scoring an average of 4.5. The AI feature that provides adaptive hints was also positively received, averaging 4.3, and participants indicated they would recommend the game to other students, with an average score of 4.5.

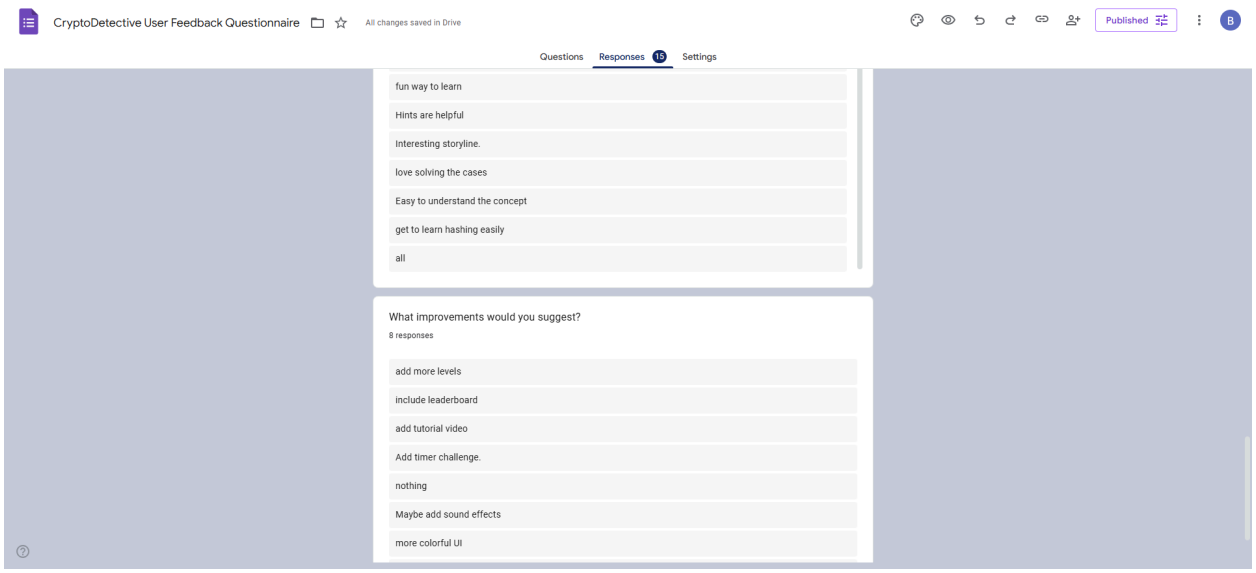

Figure 11: Qualitative User Suggestions for Prototype Improvement

Students enjoyed the detective theme, storyline, and interactive case-solving, which made learning cryptography fun and engaging. Many found the AI hints helpful for understanding concepts like hashing, digital signatures, and encryption without giving away answers. The interface was simple and easy to use, allowing students to focus on solving the cases. Suggestions for improvement included adding more levels, cases, visuals, sound effects, tutorial videos, timers, or leaderboards to make the game even more engaging. Overall, the user testing showed that CryptoDetective effectively teaches cryptography in an interactive way, while the feedback offers ideas for future improvements in design, gameplay, and learning support.

12.0 Individual Reflections

12.1 Ain Najiha Binti Junaidi

In this project, I started initiating the team by suggesting the title and outline the cryptographic concepts during the proposal phase. After receiving a few feedback and agreements from the members, I started defining the game concept to the whole team to ensure everyone is clear and they are able to build the game based on the concepts and scope that have been set. My input inspired the team straight away to set the project's direction and build the foundation idea. Later, I finalize the cryptography concepts section during the documentation phase by writing the detailed explanations using a little research and the knowledge I have.

This project equipped me with few skills in multiple areas. Firstly, I sharpened my research abilities by diving into cryptography resources and learning to simplify complex ideas for age-appropriate analogies. For instance, it is far easier to describe basic analogies by comparing and analyzing keys than explaining complex mathematical procedures. Other than that, I was able to improve my collaborative and communication skills through real-time brainstorming to facilitate discussions to align on project goals and features. Besides, generating ideas with game concepts and designs that fit the project's objectives has improved my ability to solve creative problems. Even though I was not in charge of developing the prototype, I also gained hands-on experience by being able to review the web development process.

Lastly, several challenges I faced during this project were adapting to the target audience as they needed to be exposed to terms that were easy to understand and comprehend to prevent them from being overwhelmed. Moreover, aligning ideas can be difficult as every member can have different types of opinion and ideas for this project. Organizing and coordinating the tasks also can be puzzling since we need to identify each strengths and weaknesses of each member to ensure the project production can be completed according to the deadlines. Hence, this project emphasized to me how to overcome these challenges through continuous collaboration and communication, adaptability and creative problem solving.

12.2 Balqis Nabilah binti Muhammad Azwan

Throughout this project, I actively contributed to the development of the problem statement, learning outcomes, and user testing components. I was responsible for identifying the key challenges faced by secondary school students in understanding cryptography concepts and translating these issues into a clear problem statement. In addition, I helped define learning outcomes that focused on conceptual understanding rather than mathematical complexity, ensuring the content was age-appropriate. I also played a significant role in conducting user testing by preparing the feedback questionnaire, collecting responses from participants, and analyzing the results to evaluate usability, engagement, and the effectiveness of the AI hint feature.

This project allowed me to develop and strengthen several skills. I improved my ability to communicate technical concepts in a simplified and user-friendly manner, particularly when explaining cryptography ideas such as message integrity and authenticity. I also gained experience in user experience evaluation by analyzing feedback using Likert-scale responses and open-ended questions. Furthermore, working on this project enhanced my teamwork and communication skills, as continuous collaboration and idea-sharing were required to ensure consistency across the project documentation.

One of the main challenges I faced was balancing technical accuracy with simplicity. Cryptography concepts can be complex, and ensuring they were explained correctly without overwhelming students required careful consideration. Another challenge was interpreting user feedback from a relatively small sample size, which required thoughtful analysis to draw meaningful conclusions. Additionally, coordinating tasks and aligning ideas within the group took time and effective communication. Despite these challenges, the experience helped me better understand teamwork, adaptability, and the importance of user-centered design in educational applications.

12.3 Mior Danial Hakim Bin Mior Zainul Basri

My contribution to the project was coming up with the game flow and overall layout. I took part in coming up with the game mechanics and writing out the different case-based missions where players have to analyze messages to determine if they are real or modified. I also made sure the game presents cryptography concepts such as hashing, digital signatures and encryption in a simple and easy-to-understand way for secondary school students. I executed the game through coding in HTML, CSS and JavaScript, thus giving life to design and cases.

During the course of this project, I got the chance to learn practically in designing educational games and planning interactive learning activities. Through the coding process of the game, I was able to improve my coding skills and learned how to apply real-world cryptography concepts like hashing and digital signatures in the simplest way and portray it in the game. Here, I got the chance to practice the skill of communicating technical concepts clearly and simply which improve my communication and teaching skills. Working closely with my groupmates, I got the chance to practice teamwork, coordination and problem-solving in a group since we had to generate ideas, share tasks and refine the game so that our final prototype was consistent, engaging and easy to use.

Some of the challenges I faced were showing cryptography concepts in a fun and understandable way through the game without losing its meaning. It was also difficult to make the game engaging while keeping it simple enough for secondary school students since it had to be difficult but not confusing or frustrating at the same time. I overcame these challenges through careful planning, testing and working together with my group, which allowed us to refine the game and create a final prototype that worked well and achieved our learning goals.

12.4 Wan Muhammad Afif Zikry bin Wan Ruslan

My contribution to the project involved collaborating on the design of the AI logic and educational content for the CryptoDetective cases. Together with Mior, I took part in developing the Condition–Action rules that control when the AI assistant appears and what type of hints it provides based on the player's mistakes, such as confusing hashing with encryption. I helped identify common student errors and co-wrote instructional guided support hints that guide players without directly giving away answers. I also contributed to the AI architecture section of the technical documentation and worked closely with the team to ensure the AI logic aligned well with the implemented code.

During the course of this project, I learned how to apply instructional scaffolding in an educational game by balancing guidance and challenge. I gained a better understanding of how rule-based AI systems work and how they differ from generative AI, especially in using structured logic to support learning. I also improved my ability to explain complex cybersecurity concepts such as hashing, encryption, and digital signatures in a simple way that is suitable for secondary school students. By working closely with other group members, I strengthened my communication skills and learned the importance of coordinating design ideas with technical implementation.

Some of the challenges I faced included designing AI hints that were helpful without being intrusive, as it was difficult to predict all possible student mistakes and ensure the hints supported learning without reducing problem-solving. Another challenge was clearly separating the concepts of hashing and encryption, which are commonly confused by beginners. I overcame these challenges through multiple revisions of the hint logic and dialogue, careful use of clear and precise terminology, and continuous collaboration and feedback from my teammates.